

An Accessible, Elastomer-Based Magnetic Tactile Sensor for Robotic Fingertips

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Abstract—We present an accessible elastomer-based magnetic tactile sensor designed for robotic fingertips and dexterous manipulation. The sensor combines a soft compliant silicone structure with magnetic field sensing to capture contact-induced deformation and infer external loading. To build a data-driven force estimator, we collect synchronized tactile and force torque measurements during controlled calibration motions across multiple contact depths, start points, and motion directions. A multilayer perceptron is then trained to map flattened tactile readings to three-dimensional force outputs. The resulting model enables real-time force prediction from raw magnetic measurements and provides a practical sensing pipeline for robotic grasping and manipulation. Experimental results show that the proposed approach can estimate contact force components with sufficient accuracy for feedback control and tactile perception tasks while remaining simple to calibrate and deploy.

Index Terms—robotics, tactile sensing, multilayer perception, dexterous manipulation

I. INTRODUCTION

Robotic manipulation requires fingertip sensing that can provide both contact detection and force direction feedback. Without this information, grasping and in-hand manipulation are more likely to slip, saturate contact forces, or damage fragile objects. However, many commercial tactile sensors remain expensive and difficult to integrate into small research platforms, which limits their use in teaching and prototyping environments.

This work presents a low-cost magnetic tactile sensor that is sized for robotic fingertips and designed to estimate three-dimensional contact forces from raw magnetic field measurements. The sensor uses a custom printed circuit board with five magnetometers and embedded magnets covered by a compliant silicone elastomer. When an external contact deforms the elastomer, the relative position of the magnets changes and the magnetic field pattern is altered. These field changes are captured by the electronics, transferred to a host computer through an Arduino interface, and converted into force estimates by a learned model.

To calibrate the sensor, we collect a large dataset using an xArm robot equipped with a force-torque sensor and an indenter. The robot performs controlled quasi-static contacts across multiple start points, motion directions, and indentation depths, producing paired tactile readings and ground-truth

force labels. A multilayer perceptron is then trained to map the tactile readings to three force components in Newtons. This data-driven formulation avoids a complex analytical contact model while still supporting real-time inference.

The main contribution of this paper is a complete sensing pipeline for affordable robotic fingertips, including sensor hardware, synchronized calibration data collection, and learned three-dimensional force estimation. The resulting system is intended for dexterous manipulation tasks that require practical, repeatable tactile feedback in a compact and low-cost form factor.

II. RELATED WORK

A. Soft Tactile Sensing in Robotics

Soft sensing skins have seen significant interest in recent years because they enable the passive conformal contact required for stable grasping and dexterous manipulation. A variety of sensing principles have been explored to capture contact dynamics, including resistive, capacitive, and piezoelectric networks. While effective in certain form factors, resistive and capacitive soft skins often require direct electrical connections routed into the elastomer itself. This requirement complicates fabrication and frequently leads to integration challenges or early hardware failure as the soft material degrades over repeated dynamic interactions [2], [3].

B. Vision-Based Tactile Sensors

To achieve high-resolution contact data, vision-based tactile sensors (such as GelSight [4] and DIGIT [5]) track the internal deformation of an illuminated elastomer using embedded cameras. These sensors excel at capturing rich spatial geometries and shear forces. However, optical systems inherently require a clear line of sight between the camera and the contact surface [6]. They are constrained by the camera's depth-of-focus, which places a hard limit on the minimum distance to the elastomer. Consequently, these sensors are often too bulky or expensive to integrate into small research platforms or tight multi-fingered robotic workspaces.

C. Magnetic Tactile Sensors

Magnetic sensing offers a compelling alternative by measuring applied deformations as changes in magnetic flux recorded by nearby magnetometers [1], [9]. By utilizing magnetic

microparticles or discrete embedded magnets within a soft matrix, this approach physically decouples the electronic circuitry from the passive elastomer. This separation not only allows for a highly compact footprint—ideal for robotic fingertips—but also dramatically simplifies maintenance, as the passive interface can be cheaply replaced without rewiring [9].

A notable recent advancement in this domain is ReSkin, which utilizes a magnetic elastomer skin and magnetometer arrays paired with multi-sensor machine learning to provide versatile, replaceable tactile sensing [1]. While approaches like ReSkin utilize self-supervised calibration to adapt to new skins across large surface areas, our work focuses specifically on fingertip-scale force estimation. We demonstrate that a highly accessible fabrication process paired with a straightforward multilayer perceptron can effectively map raw magnetic readings directly to three-dimensional forces. This avoids the need for complex analytical contact models or extensive multi-sensor calibration pipelines [7], [8], providing a practical, real-time sensing solution optimized for low-cost robotic integration.

III. SYSTEM DESIGN

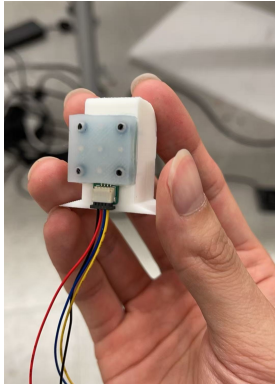


Fig. 1. The tactile sensor mounted on a 3D-printed fingertip.

A. Fabrication

1) *Silicone Elastomer*: We use Smooth-On Dragon Skin 10 Slow polymer (1:1 mix by weight) for the elastomer layer. Mix thoroughly for 2–3 minutes and degas in a vacuum chamber if available. Pour the mixed silicone into 3D-printed molds and place five small N50 disc magnets (D1001-10, 1.50 mm dia × 0.50 mm th) into the mold so that, after curing, each magnet is centered approximately 1 mm above the corresponding magnetometer on the PCB. Cure at room temperature for a minimum of 4 hours; overnight curing is recommended for consistent mechanical properties. After demolding, trim any flash, inspect for voids or bubbles.

2) *Circuit Board*: The PCB hosts five MLX90393 magnetometers placed to match the elastomer’s magnet positions. Each sensor has a local decoupling capacitor; the board includes the pull-ups resistors and a 4-pin JST-SH connector for power and the I2C communication lines. Two assembly variants are supported: (1) a single-board five-sensor layout

using hardware address changes or an I2C multiplexer(MUX) when address conflicts occur, and (2) a multiplexed setup that uses a TCA9548A to route multiple identical I2C devices. Align the cured elastomer above the PCB, verify magnet-to-sensor alignment, and secure the stack with four M2 screws through the alignment holes.

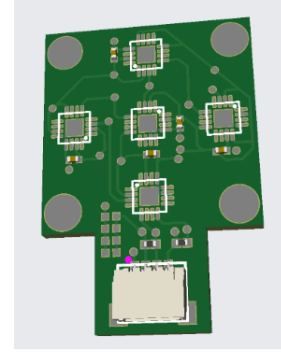


Fig. 2. The PCB design.

B. Communication

Data transmission is handled via the I2C protocol. For single-sensor configurations, the system transfers 20 continuous data streams (temperature, B_x , B_y , and B_z for each of the five magnetometer chips) to an Arduino UNO microcontroller at a sampling rate of approximately 2000 Hz. The microcontroller processes these signals and transmits them over a USB serial connection to a host PC.

To scale the system for multi-sensor configurations, an I2C MUX is utilized to route the individual sensor lines into a single I2C bus. In this expanded setup, the microcontroller is upgraded to an Adafruit QT Py to accommodate the increased memory and processing requirements.

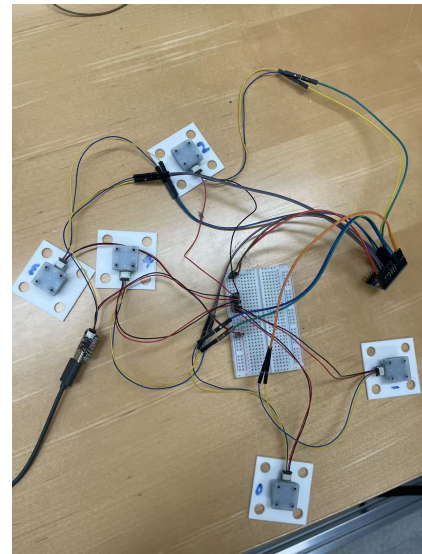


Fig. 3. Multiple sensors wiring.

IV. EXPERIMENT SETUP

A. Hardware Apparatus

The experimental calibration arena is constructed on a rigid aluminum optical breadboard to ensure structural stability and geometric repeatability. The custom magnetic tactile sensor is rigidly secured to the breadboard via a 3D-printed fixture mount. The sensor's onboard electronics are interfaced with an Arduino microcontroller, which streams raw data over a USB serial connection to a host PC.

For controlled physical interactions, a 7-axis robotic arm (xArm 7, UFactory) is securely mounted onto the same optical breadboard. A high-precision three-axis force-torque (F/T) sensor (XJC 6-axis) is attached to the robot's end-effector to record ground-truth contact loads. Finally, a custom 3D-printed indenter is mounted directly to the active sensing face of the F/T sensor to serve as the physical contact interface.

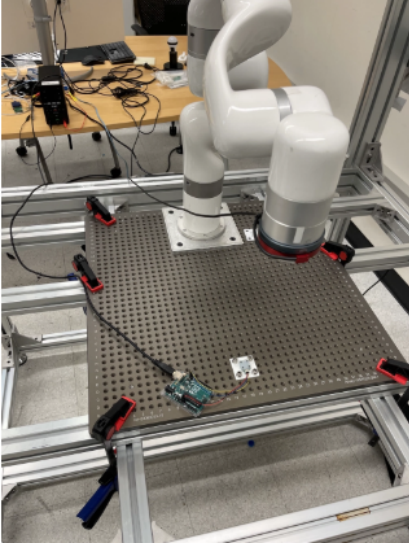


Fig. 4. Experimental setup for automated tactile sensor calibration.

B. Data Collection

The automated data collection routine evaluates a diverse range of contact scenarios across normal and shear axes. The calibration procedure comprises a combinatoratoric sweep across multiple parameters:

- **Spatial Entry Points:** A grid of 9 distinct start points uniformly distributed across the elastomer surface.
- **Motion Directions:** For each entry point, the robot actuates the indenter along 8 distinct directions to induce varied multi-axis shear forces.
- **Indentation Depths:** Continual normal compressions ranging from 0.2 mm to 1.4 mm relative to the initial undeformed elastomer envelope.

During each discrete trial, the robotic arm repositions to a designated start point, descends to establish initial touch, and executes a continuous exploration path along one of

the prescribed directions. Throughout this trajectory, time-synchronized measurements from the five onboard magnetometers and the XJC F/T sensor are captured simultaneously. The normal and shear force components (F_x, F_y, F_z) logged by the F/T sensor serve as the definitive ground-truth labels. The resulting paired vectors of raw magnetic flux variants and multi-axis forces are serialized and stored for network model training.

V. SENSOR MODEL

A. Magnetic Field to Force Mapping

The tactile sensor produces raw magnetic flux measurements that encode the elastomer deformation induced by external contact. The relationship between magnetic field patterns and applied forces is inherently nonlinear; when the elastomer compresses and shears under load, the relative positions of the embedded magnets shift continuously, producing complex changes in the field distribution observed by the sensor array.

Our approach bypasses the need for explicit analytical contact models by learning this mapping directly from calibration data. At each discrete sampling instant, the five onboard magnetometers yield three-axis magnetic flux density measurements:

$$\mathbf{B}_i = [B_{x,i}, B_{y,i}, B_{z,i}]^T, \quad i = 1, \dots, 5 \quad (1)$$

These fifteen values are concatenated into a single feature vector:

$$\mathbf{x} = [B_{x,1}, B_{y,1}, B_{z,1}, \dots, B_{x,5}, B_{y,5}, B_{z,5}]^T \in \mathbb{R}^{15} \quad (2)$$

The three-dimensional contact force is denoted as:

$$\mathbf{f} = [F_x, F_y, F_z]^T \in \mathbb{R}^3 \quad (3)$$

where F_x and F_y represent shear force components and F_z represents the normal force. Our objective is to learn a function $\Phi : \mathbb{R}^{15} \rightarrow \mathbb{R}^3$ such that $\hat{\mathbf{f}} = \Phi(\mathbf{x})$ provides accurate force estimates in real-time.

B. MLP Architecture and Training

We employ a fully-connected multilayer perceptron (MLP) to learn the mapping Φ . The network architecture is as follows:

- **Input layer:** 15 neurons (raw magnetic flux channels)
- **Hidden layer 1:** 200 neurons with ReLU activation
- **Hidden layer 2:** 200 neurons (linear)
- **Bottleneck layer:** 40 neurons (linear)
- **Hidden layer 3:** 200 neurons with ReLU activation
- **Hidden layer 4:** 200 neurons with ReLU activation
- **Output layer:** 3 neurons (force predictions for F_x, F_y, F_z)

1) *Data Preprocessing and Normalization:* Raw magnetic flux measurements are preprocessed using standardization. Prior to training, the training set features are normalized using:

$$\mathbf{x}_{\text{normalized}} = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}} \quad (4)$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ are the mean and standard deviation computed from the training data. The same scaler is applied to validation and test data, and is retained for deployment to ensure consistent scaling at inference time.

2) *Training Procedure*: We employ 5-fold cross-validation to assess generalization performance and select the final model. For each fold, the dataset is partitioned into training and validation subsets in an 80/20 split. Training proceeds using the following hyperparameters:

- **Optimizer**: Adam with learning rate $\alpha = 1 \times 10^{-3}$
- **Loss function**: Mean Squared Error (MSE)
- **Batch size**: 32 samples
- **Max epochs**: 100

The MSE loss is computed for each mini-batch and back-propagated through the network. Over the course of training, the optimizer adapts the per-parameter learning rates, allowing efficient convergence despite the relatively sparse and nonlinear structure of the sensor data.

3) *Evaluation Metrics*: Model performance is evaluated using the following per-axis and vector metrics:

- **Mean Absolute Error (MAE)**
- **Root Mean Squared Error (RMSE)**
- **Vector magnitude error**: $\|\hat{\mathbf{f}} - \mathbf{f}\|_2$
- **Accuracy within tolerance**: Fraction of samples for which $|\hat{f}_{i,n} - f_{i,n}| \leq 0.5 \text{ N}$

4) *Final Model Training*: Following cross-validation, the full dataset is used to train a final model with identical hyperparameters. This model is saved and deployed for real-time force inference on the robot platform. The learned weights, biases, and the fitted scaler are serialized and loaded during system operation.

C. Results

The 5-fold cross-validation performance metrics for the multi-axis force estimator are comprehensively summarized in Table I and Table II. The network demonstrates exceptional predictive accuracy across all three orthogonal axes, successfully capturing the highly nonlinear relationship between elastomer deformation, magnetic field distribution shifts, and external loading.

TABLE I
CROSS-VALIDATION ERROR METRICS FOR FORCE ESTIMATION (MEAN $\pm$ SD OVER FIVE FOLDS)

Force Component	MAE (N)	RMSE (N)
F_x (Shear)	0.0392 ± 0.0015	0.0540 ± 0.0025
F_y (Shear)	0.0461 ± 0.0038	0.0642 ± 0.0055
F_z (Normal)	0.0858 ± 0.0072	0.1228 ± 0.0106
$\mathbf{F}$ (Vector Magnitude)	0.1183 ± 0.0076	0.1489 ± 0.0109

TABLE II
CROSS-VALIDATION ACCURACY WITHIN 0.5 N TOLERANCE WINDOW

Target Axis	Accuracy (Fraction $\pm$ SD)
F_x within tolerance	1.0000 ± 0.0000
F_y within tolerance	0.9998 ± 0.0002
F_z within tolerance	0.9946 ± 0.0030

As indicated by the experimental results, the mean absolute error (MAE) for the planar shear force components stays remarkably low, yielding $0.0392 \pm 0.0015 \text{ N}$ for F_x

and $0.0461 \pm 0.0038 \text{ N}$ for F_y . The normal force component (F_z) exhibits a slightly higher baseline error profile of $0.0858 \pm 0.0072 \text{ N}$ MAE and $0.1228 \pm 0.0106 \text{ N}$ RMSE. This minor divergence is structurally expected, given the higher stiffness and viscoelastic resistance encountered by the embedded magnets during deep orthogonal compressions compared to lateral planar displacements.

Crucially, the total combined 3D vector magnitude error maintains a tight envelope with an MAE of $0.1183 \pm 0.0076 \text{ N}$ and an RMSE of $0.1489 \pm 0.0109 \text{ N}$. When evaluating performance against a standard operational tolerance window of $\leq 0.5 \text{ N}$, the model achieves a flawless tracking performance of 100% accuracy on the F_x axis, 99.98% on the F_y axis, and 99.46% on the normal force F_z axis. Furthermore, the tightly bounded standard deviations across all metrics demonstrate outstanding consistency across separate dataset partitions, confirming that the learned model generalizes robustly without overfitting to spatial configurations or unique loading vectors. This level of accuracy is highly sufficient for closing real-time force-feedback control loops during delicate robotic grasping and manipulation tasks.

D. Experimental Force Tracking Validation

To evaluate the generalization capability and practical deployment of the learned force estimator, we conduct an online verification experiment using an indenter in human fingertip shape that distinct from the hemispherical probe used during data collection. The xArm 7 robot executes a series of dynamic contact trajectories, during which real-time model predictions are compared directly against ground truth data streams from the XJC force-torque sensor.

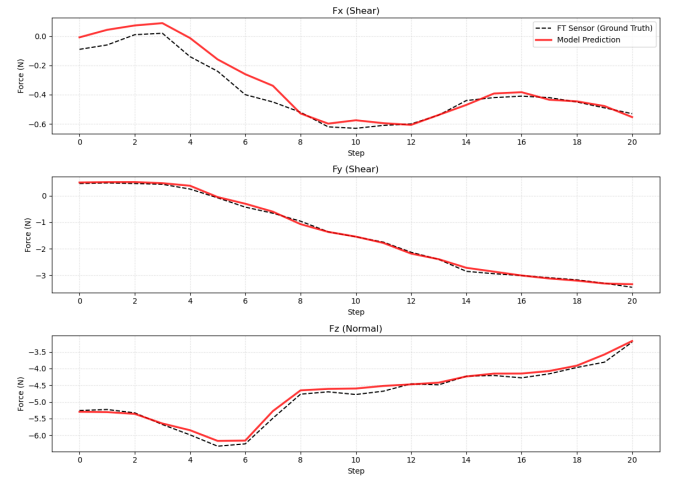


Fig. 5. Dynamic force tracking comparison for a multi-axis contact path at an indentation depth of 0.2 mm using a novel indenter.

1) *Multi-Axis Sweeps*: First, combined normal and shear loading conditions are evaluated by indenting the elastomer surface and actuating the arm continuously along a direction. This procedure is conducted at two distinct depths: a shallow

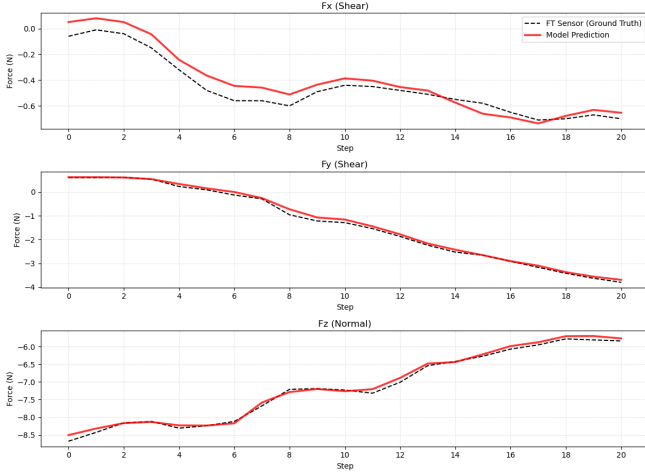


Fig. 6. Dynamic force tracking comparison for a multi-axis contact path at a deeper indentation depth of 0.5 mm.

compression of 0.2 mm (Fig. 5) and a deeper compression of 0.5 mm (Fig. 6).

As demonstrated in the 0.2 mm sweep, the MLP model tracks the complex multi-axis loading profile with high fidelity. For the shear components, the estimated F_x and F_y force tightly bounds the ground-truth curve across the entire 20-step sequence. The normal force (F_z) exhibits highly accurate baseline matching. At a deeper indentation depth of 0.5 mm, the tracking performance remains robust. Despite the modified contact mechanics introduced by the new indenter tip under higher strain, the model accurately maps the continuous rise of the shear load F_y up to 4 N and successfully captures the deeper normal force profile bounding down to -8.5 N.

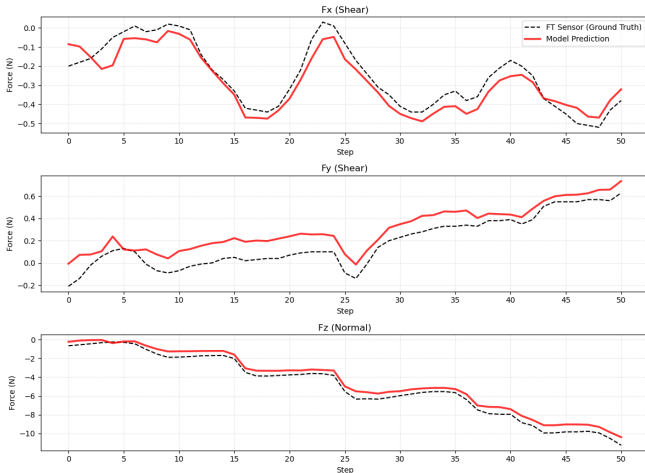


Fig. 7. Continuous normal force tracking verification during a pure vertical indentation profile up to a maximum depth of 1.2 mm.

2) *Pure Normal Loading Profile:* Second, to isolate the sensor's cross-axis decoupling characteristics, a pure vertical

indentation sequence is conducted without lateral shear actuation (Fig. 7). The robot drives the novel indenter normally into the sensor up to a maximum depth of 1.2 mm over 50 sequential steps.

The normal force (F_z) estimation results reveal near-flawless tracking of the reference profile, capturing the steep step-wise compressions down to approximately -11 N with remarkable linearity and negligible hysteresis. Crucially, during this heavy compression cycle, the model successfully maintains the shear forces close to the zero-force baseline. The F_x and F_y predictions are successfully bounded within a tight margin (0.0–0.5 N), proving that the learned pipeline effectively decouples orthogonal structural deformations from planar components without suffering from significant cross-talk or output saturation.

These validations altogether demonstrate that the proposed sensor system achieves excellent calibration transferability. The data-driven force mapping functions reliably when subjected to completely novel mechanical boundary conditions and contact shapes, confirming its suitability for closed-loop tactile perception and feedback control in unconstrained robotic manipulation.

VI. CONCLUSION

We presented an accessible, fingertip-scale magnetic tactile sensor that combines a compact multi-magnetometer PCB with a replaceable elastomeric skin and a data-driven calibration pipeline. By collecting synchronized magnetometer readings and ground-truth force-torque labels with a robot-mounted indenter, we trained a lightweight multilayer perceptron to map 15-channel magnetic flux measurements to three-dimensional contact forces in real time. Experimental results demonstrate low per-axis error, tight vector-magnitude tracking, and reliable transfer to novel indenters and loading conditions, confirming the approach's practicality for closed-loop tactile feedback in dexterous manipulation.

While the current implementation emphasizes accessibility and compactness, several directions can extend its utility and robustness. First, scaling the layout to higher-resolution sensor arrays and multi-finger skins will expand spatial coverage and enable the estimation of richer contact geometries. Second, incorporating online or self-supervised calibration techniques can mitigate the effects of sensor drift, material hysteresis, and manufacturing variability, allowing the model to adapt seamlessly to skin replacements without requiring the tedious re-collection of ground-truth data. Finally, future iterations will explore optimized elastomer compositions and mechanical mounting configurations to maximize structural durability and long-term repeatability while investigating low-cost manufacturing methods to facilitate wider deployment across diverse robotic platforms.

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